



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Club Collab

Group No : __, CSE370 Lab Section : __, Spring 2026		
ID	Name	Contribution
23201457	Araf Kamal	Frontend+Backend
23201465	Shanid sajjatuz Islam	Frontend+Backend
23201146	Md Reduyan Islam Bhuiyan	Backend

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	4
3	ER/EER Diagram	5
4	Schema Diagram	6
5	Normalization	7
6	Frontend Development	7
7	Backend Development	8
8	Source Code Repository	9
9	Conclusion	9
10	References	9

Introduction

Club-Collab is a centralized database and web application designed for BRAC University to solve two major administrative headaches: **resource booking conflicts** and **unverified student volunteer tracking**.

In brief, the project serves as an all-in-one university club management system with the following core features:

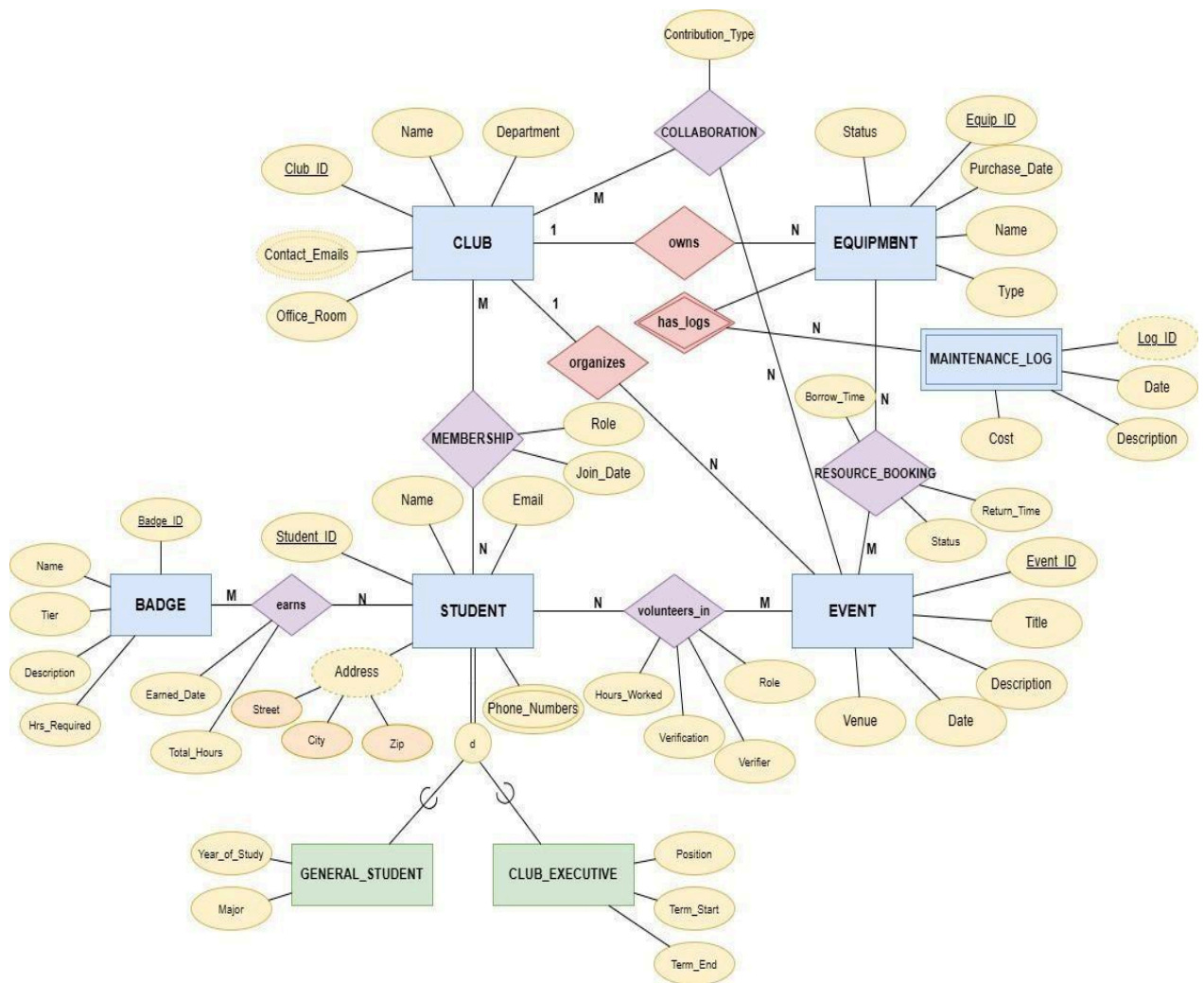
- **Conflict-Free Resource Booking:** Clubs can register and share equipment (like cameras, projectors, or microphones). The system uses database constraints to prevent double-booking and features intelligent triggers that automatically cancel future reservations if an item is reported as "Damaged".
- **Verified Volunteer Tracking & Gamification:** Students can log their volunteer hours for different events, which are then verified by club executives. To encourage engagement, the system features an automated "Badge System" that awards tiered medals (Bronze to Diamond) as students accumulate verified volunteer hours, allowing them to build a reliable portfolio of their contributions.
- **Centralized Entity Management:** It manages Clubs (including their contact info and office locations), Events (supporting multi-club collaborations), and Students (using a specialized hierarchy to differentiate between General Students and Club Executives).
- **Analytics Dashboard:** It provides admins and executives with data-driven insights, such as equipment utilization ROI, top cross-club volunteers, and event success metrics.

Ultimately, Club-Collab takes chaotic, manual club processes and transforms them into an efficient, automated, and data-driven platform.

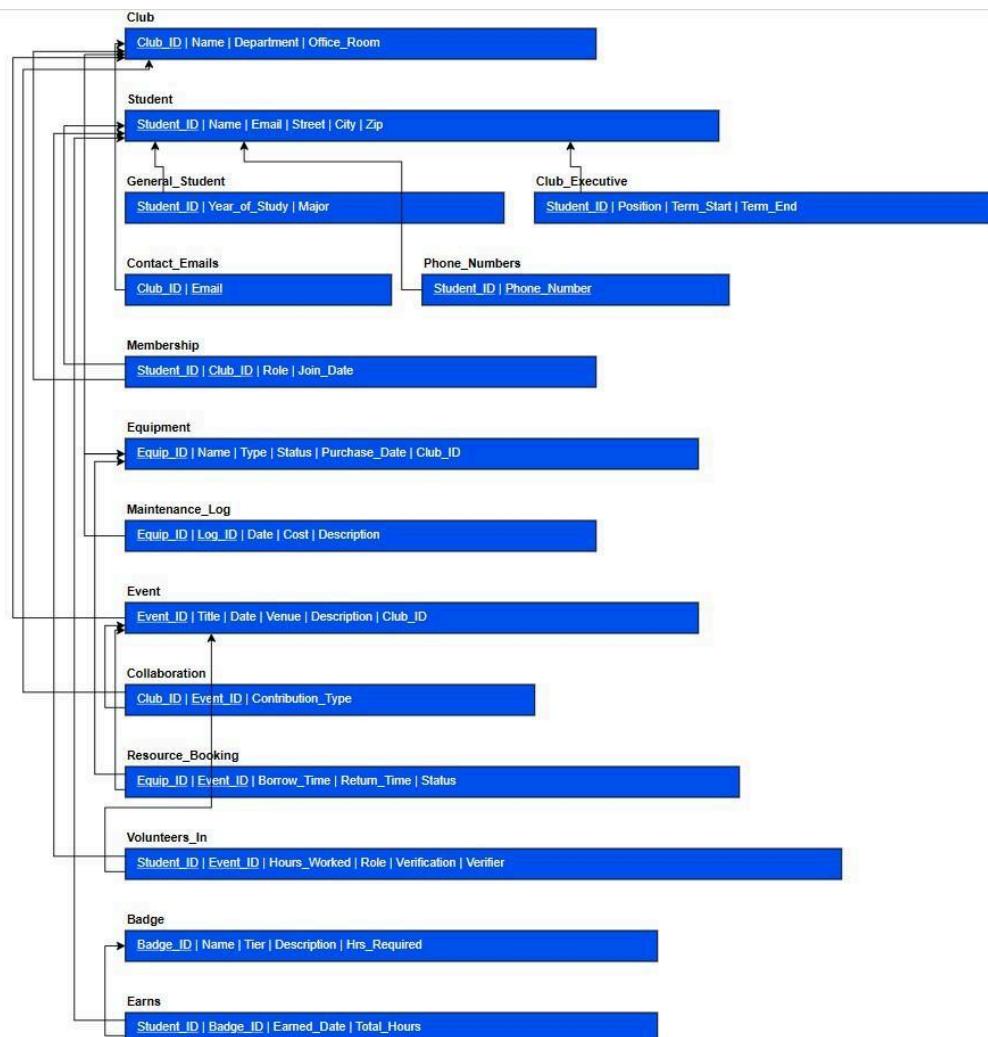
Project Features

ID, Name	Features [3 per member]	
23201457, Araf	Ft 1	Events
	Ft 2	Volunteers
	Ft 3	Badges/Analytics
23201146, Reduyan	Ft 1	clubs
	Ft 2	Database connection
	Ft 3	Memberships
23201465, shanid	Ft 1	Volunteer Tracking
	Ft 2	Equipment
	Ft 3	Bookings

ER/EER Diagram



Schema Diagram



Normalization

- Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

The database schema is fully normalized up to the Third Normal Form (3NF).

1NF (Atomicity): The schema achieves 1NF by ensuring all attributes contain atomic values. For example, the student address is decomposed into Street, City, and Zip, and multi-valued attributes like Phone_Number and Email are moved to separate tables.

2NF (Full Functional Dependency): The schema is in 2NF because every non-prime attribute is fully dependent on the entire primary key. In junction tables like Membership or Resource_Booking, attributes like Role or Status depend on the composite key (the combination of both IDs) rather than just one part of it.

3NF (No Transitive Dependencies): The schema reaches 3NF because all non-prime attributes depend solely and directly on the primary key. There are no instances where a non-key attribute depends on another non-key attribute.

Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23201457, Name : Araf Kamal

- Login**
- Index**

Contribution of ID : 23201465, Name : Shanid Sajjatuz Islam

- Bidding**
- Sign up**

Contribution of ID : 23201146, Name : Md Reduyan Islam Bhuiyan
Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23201457, Name : Araf Kamal

Engagement, Bidding & Analytics

- crud_volunteers.php: The CRUD engine for logging volunteer hours.
- get_volunteers.php: Retrieves the volunteer logs for verification.
- volunteers_api.php: Advanced filtering for volunteer portfolios.
- badges.php: Logic for the leaderboard and calculating earned achievements.
- badges_api.php: Administrative management of badge requirements/tiers.
- bidding_api.php: Implements the inter-club equipment bidding economy.
- analytics.php: Generates high-level reports via stored procedures.
- views.php: A centralized gateway to access the 11 pre-defined SQL Views.
- search.php / search_api.php: Global search logic across equipment and events.

Contribution of ID : 23201465, Name : Shanid Sajjatuz Islam

Physical Logistics (Resources & Events)

- equipment.php: The main controller for registering and updating equipment.
- get equip.php: Fetches the inventory with owner club details.
- crud_bookings.php: The engine for creating and managing resource reservations.
- get_bookings.php: Read-only access to the booking schedule.
- bookings.php / bookings_api.php: Logic for status updates and conflict reporting.
- maintenance.php: Handles the Weak Entity logs for equipment repairs.
- events.php: Manages event lifecycles using SQL transactions.
- get_events.php: Summarizes event statistics and collaboration counts.

Contribution of ID : 23201146, Name : Md Reduyan Islam Bhuiyan

Operations & Identity Management

- students.php: Main CRUD for student records (handling ISA specialization).
- get_students.php: Specialized retrieval of student profiles with club roles.
- students_api.php: API endpoints for student data management.
- auth.php: Authentication logic and role verification (Executive vs. General).
- get_clubs.php: Aggregates club information and contact emails.
- clubs_api.php: CRUD operations for club profiles and locations.
- memberships.php: Manages the relationships between students and clubs.
- db_connect.php: The central database connection configuration.
- system_health.php: Checks connectivity and status of database constraints.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

<https://github.com/perashanid/club-collab.git>

Conclusion

The development and architecture of Club-Collab, a web-based management system designed to streamline university club operations. The project utilizes a PHP backend with MySQLi Procedural scripts to handle tasks such as user authentication, resource booking, and volunteer tracking. Key features include a gamified badge system for volunteers and a conflict-free equipment reservation tool supported by automated database triggers. The documentation also covers technical troubleshooting, including database connectivity fixes and the transition from PDO to procedural code. Collectively, these materials offer a comprehensive overview of how structured data and API endpoints facilitate efficient club collaboration and administrative oversight.

References:

https://www.w3schools.com/php/php_intro.asp

https://apps.bux-home.bracu.ac.bd/learning/course/course-v1:buX+CSE370+2025_Fall/block-v1:buX+CSE370+2025_Fall+type@sequential+block@52dd4ee286aa465fa5f6a3f93c5976bb/block-v1:buX+CSE370+2025_Fall+type@vertical+block@6389c87707764b57a80ab859bc0f503c

https://www.youtube.com/watch?v=nu_pCVPKzTk